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Influence of slope on root system anchorage of *Pinus yunnanensis*

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ABSTRACT

In southwestern China, Yunnan pines (*Pinus yunnanensis*) have been extensively cultivated on barren hills for reforestation and ecological engineering. The objective of this work is to study the influences of slope gradient on anchorage of root systems of *P. yunnanensis*. Pulling experiments were carried out at low (5–6°), moderate (25–26°) and high gradient (42–43°) by using selected 15-year-old *P. yunnanensis*. The results showed that the anchorage resistances induced by the slope gradient were significantly ($P < 0.01$) different from each other and followed in the order of high slope > moderate slope > low slope. Anchorage strength of upslope-grown roots increased with size and length of first-order lateral roots as well as the number of second-order lateral roots. Contributions of upslope-grown roots for preventing plants from overturning varied considerably between those from different slopes. The contribution of roots growing on upslope side on high slope to the anchorage of root system reached 50%, whereas those on moderate and low slopes were about 44% and 37%, respectively. It was concluded that the anchorage resistance was closely related to slope gradient and root distribution, while upslope-grown roots were positively related to anchorage resistance.

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1. Introduction

It has been widely believed that anchorage is one of the two main functions of root systems (Coutts, 1983, 1987; Coutts et al., 1999) and it plays a vital role in preventing plants from overturning. Moreover, plant roots help stabilize forested slopes by providing root strength and modifying soil water regime (Burroughs and Thomas, 1977; O'Loughlin and Watson, 1979; Abe and Ziemer, 1991a,b; Watson et al., 1995), while the root system of riparian vegetation stabilizes streambanks by increasing tensile strength in the soil (Mallik and Rasid, 1993). The anchorage of roots and the improvement of slope stability mainly depend on the properties of root systems such as the root distribution and tensile strength (Nicoll and Ray, 1996; Stokes and Guitard, 1997; Stokes et al., 1998; Normaniza and Barakbah, 2006; Li et al., 2007), as well as soil conditions. For example, wet and heavy soils provide more resistance to uprooting than dry and sandy soils do (Ennos, 1990; Goodman and Ennos, 1999).

It has been reported that root systems not only are able to respond locally to mechanical stimulation, but also lead to modification of root traits to improve anchorage strength

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(Goodman and Ennos, 1996, 1997, 1999). For example, in the case of unidirectional wind stimulation (Fitter and Ennos, 1989), where the mechanical stimulus was not symmetrically distributed on the root systems, a close relationship of root system resistance moment with root number (Duparque and Pellerin, 1994; Stokes et al., 1996; Guigo and Herbert, 1997), as well as the changes in root angle (Crook and Ennos, 1994) and diameter (Herbert et al., 1992) were observed. Slope is a complex environmental condition that imposes several mechanical stresses to plants such as the turning moment induced by the combination of the inclination and the weights of the stem and the soil. Therefore, it is believed that anchorage strength can be improved by a modification of roots under slope condition. The mechanical characteristics of root system of plants growing on a plane land have intensively been studied (Gilroy et al., 2001). Also, the changes in root architecture resulting from slope condition (Di Iorio et al., 2005) and the influence of slope on root system development (Chiatante et al., 2003) have been explored. However, little attention has been paid to the mechanical characteristics of the root system in plants growing on slopes, especially the anchorage of root system of plants growing slope have never been thoroughly investigated. The purpose of the present study is to investigate the influence of slope gradient on anchorage of plant root systems and identify the anchorage contribution by different directional growing roots.

2. Materials and methods

2.1. Study site

The study sites are located in the Field Observation Station for Runoff and Erosion in the Ginping Gorge (1341–1376 m altitude, 28°32′N, 101°38′E), southwestern China. The local climate is subtropical monsoon with a mean annual temperature of 17.1 °C and average annual precipitation of 1084.6 mm. The mean monthly temperature ranges from 22.6 °C in July (warmest) to 9.3 °C in January (coldest).

The Station, established in 1991, has three types of slopes, each of which accounts for about 1200 m^2. Those slopes have same aspect of SW 9° and different gradient of 5–6° (defined as low slope), 25–26° (defined as moderate slope) and 42–43° (defined as high slope). *Pinus yunnanensis* was planted with span of about 5 m after the Station was established. There are two reasons for selecting this area to conduct our study: the first is that trees of *P. yunnanensis* were managed carefully after grown, including soil nutrient analysis every month, soil water measurement every 10 days, and management of fertilization and irrigation according to different slope conditions. These may eliminate the influence of soil condition on root systems anchorage; the second is that these sites were situated at lee slope in the gorge, and this may decrease the effect of wind loading on root systems anchorage.

Table 2 – Different treatment methods in the preparing tests

Test	Method
Control (T1)	Without any treatment
Treatment (T2)	An entrenchment was excavated and in which downslope-grown lateral roots were cut, while upslope-grown roots were preserved. Size of the entrenchment as in Fig. 1(A)
Treatment (T3)	An entrenchment was excavated and in which upslope-grown lateral roots were cut, while downslope-grown roots were preserved. Size of the entrenchment as in Fig. 1(B)

P. yunnanensis was chosen to investigate anchorage of root system, because it has been widely applied for ecological engineering in the southwest of China, and it has a total planting area over 3 million ha by an increase of more than 0.3 million ha per year (Forest Survey report of China, 2005). Pull-out tests were conducted on slopes with different gradients and pull-out resistance of individual root was tested.

2.2. Preparing tests

Preparing tests were conducted from 10 to 18 May 2006. A total of 15 plants were selected randomly from each slope gradient, and their height and diameter at breast height (DBH) were measured (Table 1), and those trees were subjected to three different treatments according to the method described by Crook and Ennos (1996) (Table 2). A total of 45 plants were selected to perform our study. After all preparing tests were finished, the entrenchments were filled with soil to recover the original compact in order to reduce the effect of movement of rotation center during pull-out process on anchorage measurement (Crook and Ennos, 1996).

2.3. Pull-out tests

Pull-out tests were conducted from 15 August to 30 September 2006. All of the winched trees were pulled down toward downslope. Before pull-out, winched trees were cut and remained only a trunk of about 100 cm aboveground, because the canopy weight would impact on the accuracy of the results and lead to inexact measurement (Cucchi et al., 2004). A motorized winch (DF2000-50, Shanghai HengJi electrical machinery factory, China) with a maximum capacity of 50 kN

Table 1 – Height and diameter at breast height (DBH) of selected *Pinus yunnanensis* on low, moderate and high slopes

	Low slope	Moderate slope	High slope	*n*	*P*
Height (m)	11.22 ± 0.27	11.16 ± 0.36	11.07 ± 0.24	15	0.25
DBH (cm)	12.12 ± 0.27	12.07 ± 0.11	12.02 ± 0.21	15	0.57

Data are means ± 1 S.E. Sample size (*n*) is the number of tested plant. *P* values are the result of multiple comparisons.

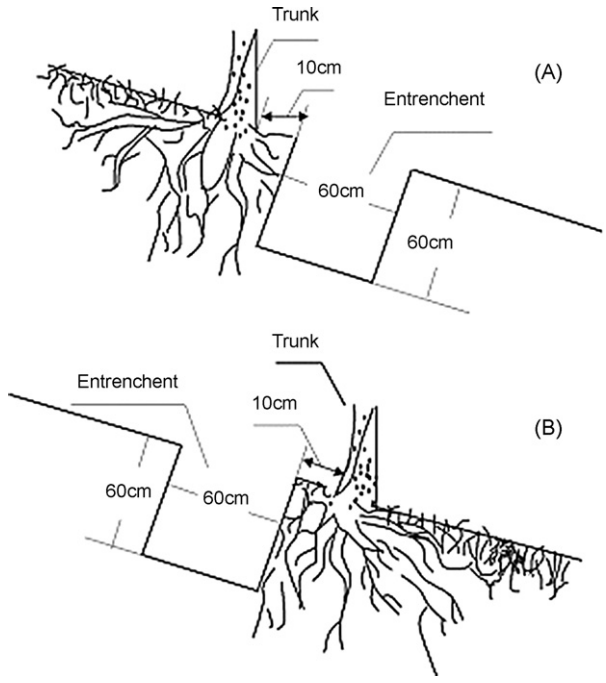


Fig. 1 – Schematic diagram of the preparing test. (A) The treatment in T2, cutting the downslope root system in entrenchment. (B) The treatment in T3, cutting the upslope root system in entrenchment.

was used to fixed on a 5 t tractor locating flat land, the winch connect winched tree by a cable that parallel to the slope surface (Fig. 2). The height of the fixed cable attachment was 50 cm above the soil surface for all trees. This height was usually lower than the one-third to half of the plant height when investigated wind loading (Moore, 2000; Peltola et al., 2000). As studying root anchorage, the attachment height should be low enough on the stem to induce root failure instead of stem breakage (Cucchi et al., 2004). The tension applied to the winched plant was measured with a transducer (TJL-1S, Shanghai TM Automation Instruments Co. Ltd., China) that was fixed between the winch and the winched tree; the data were recorded every second with a digital logger (Almemo 2290-8, Ahlborn, Germany). To determine the inclination angle α of the stem during the winching test, one inclinometer was tied to the base of the stem. The degree of inclination angle was recorded every second with another digital logger. Those

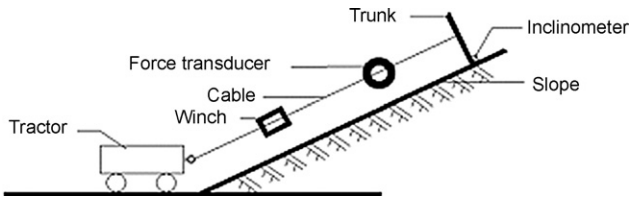


Fig. 2 – Schematic diagram of the pull-out test. Trees were winched along downslope and the force necessary was measured using a force transducer located between the winch and anchored tree. An inclinometer was attached at the stem base in order to measure stem deflection during winching.

two recorders work synchronically and simultaneously. To prevent the cable slipping away form the trunk, the bark was peeled away, the stem was packed by asbestos then it was clipped by one adjustable metal gripper. The soil layer of slope was watered till saturated and then plants were pulled down 24 h later. This way, each plant was winched over at a rate of 10 min^{-1}. Simultaneously, attention was paid to the movement of the roots, the rotation center of the trunk, and broken sound of roots. Then, the maximum overturning moment of all winched trees was calculated. In addition, the maximum inclination angle of the trunk caused by the overturning moments that reached maximum value in the T1, as a basic, was used to calculate the overturning moments of winched trees in T2 and T3 when they reached the same angle, and to weight the contributions of upslope and downslope lateral roots for plant anchorage, respectively.

2.4. *Measurements of root system and tensile strength*

The parameters of upslope- and downslope-grown root systems, such as the number, length and average diameter (Crook and Ennos, 1998), were measured. The maximum and minimal diameters of roots from 10 cm of the offshoot point were also measured by a vernier caliper. The first- and second-order lateral roots in both upslope and downslope were selected to measure tensile force by electric pull machine (KDIII, Shenzhen KaiQiangLi Test Apparatus Co. Ltd., China). Tensile strength was calculated by following formula:

$$P_g = \frac{4F}{\pi D^2}$$

where P_g (MPa) is tensile strength, F_{max} (N) the tensile force and D (mm) is the diameter of root.

Roots with growing directions parallel to the cable were selected as much as possible. The root that uses to measure tensile resistance was 10 cm in length, it was intercepted after from the offshoot point about 10 cm, each lateral root used only one sect (Zhou et al., 1999).

2.5. *Soil measurements*

Ten sampling sites were selected randomly on each slope. Soil bulk densities were measured by ring samplers of 100 cm^3 from 5–10 to 40–50 cm in depth, respectively. Soil moisture contents were measured by oven-dry method (dried at 105 °C to constant weight). Soil samples were air-dried and cleaned out the roots or other impurities and then crushed to pass a 0.149-mm sieve. Total nitrogen was determined by Kjeldahl method (McGeehan and Naylor, 1988), total potassium was determined by flame photometry method (Shi and Bao, 1980), and total phosphorus was measured by sodium carbonate (Na_2CO_3) fusion-molybdenum blue photometric method (Syers et al., 1967).

2.6. *Data analysis*

Data were analyzed with SPSS11.0 statistical software. Analysis of variance (ANOVA) was tested statistical significance of those data. Tukey test was used to compare differences among treatments, including the parameters of root systems, anchor-

Table 3 – Physical and chemical properties of soils in selected sites

Soil depth (cm)	Property	Low slope	Moderate slope	High slope	*n*	*P*
5–10 cm	Bulk density (g/cm^3)	1.11 ± 0.14	1.12 ± 0.127	1.12 ± 0.11	10	0.57
	Soil moisture (%)	22.86 ± 0.61	22.43 ± 0.73	22.21 ± 0.65	10	0.21
	Total K (g/kg)	7.86 ± 0.11	7.71 ± 0.12	7.64 ± 0.23	10	0.18
	Total N (g/kg)	2.49 ± 0.21	2.48 ± 0.18	2.47 ± 0.16	10	0.12
	Total P (g/kg)	2.28 ± 0.16	2.26 ± 0.12	2.27 ± 0.16	10	0.33
40–50 cm	Bulk density (g/cm^3)	1.33 ± 0.17	1.34 ± 0.15	1.34 ± 0.16	10	0.48
	Soil moisture (%)	18.46 ± 0.52	18.41 ± 0.54	18.33 ± 0.51	10	0.31
	Total K (g/kg)	4.12 ± 0.06	4.11 ± 0.09	4.10 ± 0.11	10	0.19
	Total N (g/kg)	1.64 ± 0.14	1.63 ± 0.12	1.62 ± 0.15	10	0.26
	Total P (g/kg)	1.48 ± 0.18	1.47 ± 0.16	1.46 ± 0.16	10	0.13

Data are means ± 1SE, *P* value shows the results of analysis of variance. Sample size is the number of selected site at each slope.

age resistance of root systems, as well as tensile resistance of lateral roots.

3. Results

3.1. *Soil property*

The upper soil layers of all slopes always have a higher nutrient and water contents and lower bulk density (Table 3). Within the same soil layer among slopes, soil properties, including total nitrogen, total phosphorus, total potassium, bulk density and water moisture had no significant difference (all $P \geq 0.12$).

3.2. *Growth and distribution of root system*

Plants selected have one taproot and several strong first-order laterals, and only a few of sink roots with a small diameter could reach the depth of 1 m under surface soil.

The amount of the first-order lateral roots was significantly different between slope gradients ($P < 0.01$) (Fig. 3A). For example, the number of first-order lateral roots in the high slope was 6.8 ± 0.63, lower than other two types of slopes, but their diameter and length were significantly higher than ones in both the moderate and low slopes ($P < 0.01$) (Fig. 3B). The length and diameter of the second-order lateral roots had no difference among three slopes ($P = 0.12$) (Fig. 3C), but the amount on the high slope was noticeably higher than others ($P < 0.01$) (Fig. 3A).

Amount and diameter of roots growing in different direction showed a considerable difference. In high and moderate slopes, the amount and diameter of the first-order lateral roots in upslope were significant higher than ones in downslope ($P < 0.01$) (Fig. 4A). The amount of the second-order lateral roots in upslope exceeded one in downslope as well ($P < 0.01$), but their diameters had no marked difference among three type slopes ($P = 0.23$) (Fig. 4D). For low slope, both the amount and diameter of roots had no difference between the upslope and downslope (all $P \geq 0.07$) (Fig. 4).

3.3. *Tensile resistance*

The tensile resistances of roots were related to location where they grow. On high and moderate slopes, the tensile resistances of the first- and second-order lateral roots in upslope were marked greater than that of downslope ($P < 0.01$), but without difference between the upslope and downslope for the low slope ($P > 0.45$) (Fig. 5).

3.4. *Anchorage and breakage of root system*

In the control test (T1), the maximum overturning moments had statistics significance ($P < 0.01$) among different slopes (Table 4), and followed in the order of high slope > moderate slope > low slope. The inclination angle showed significant differences among slope gradients when overturning moment reached a peak value. In the T2 and T3, the maximum overturning moments of the winched trees in the high slope were

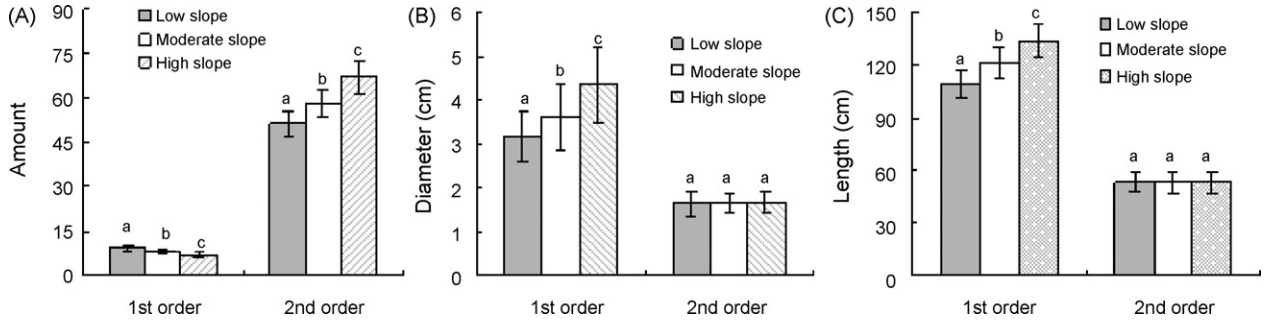


Fig. 3 – (A) Amount, (B) diameter and (C) length of the first- and second-order roots among three types of slopes. Data are means ± 1 S.E., *n* = 10 plants per slope. For each parameter, means with the same letter are not significantly ($P > 0.05$) difference.

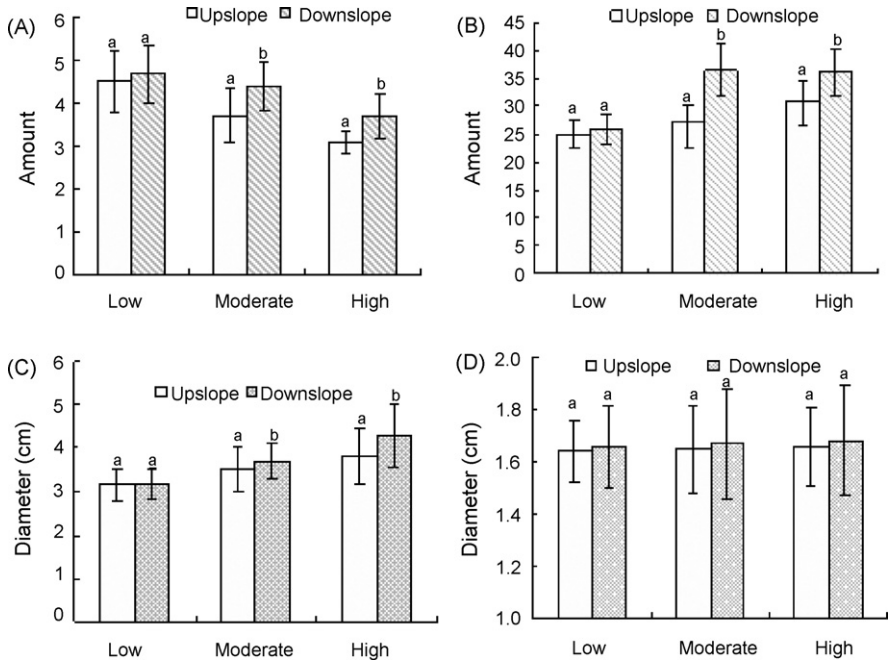


Fig. 4 – Comparison of the parameters of root system between the upslope and downslope. (A) and (B) The amount of the first- and second-order lateral roots, respectively. (C) and (D) Diameter of the first- and second-order lateral roots, respectively. Data are means ± 1 S.E., $n = 10$ plants per slope. For each parameter, means with the same letter are not significantly ($P > 0.05$) difference.

Table 4 – Maximum overturning moment (MOM) and inclination angle (IA) of selected *P. yunnanensis*

Test		Parameter	Low slope	Moderate slope	High slope	n	P
T1	(1)	MOM (kNm)	13.28 ± 0.64 A	14.13 ± 0.71 B	15.09 ± 0.68 C	5	<0.01
	(2)	IA (°)	26.8 ± 1.52 A	24.6 ± 1.67 B	22.1 ± 1.86 C	5	<0.01
T2	(3)	MOM (kNm)	10.11 ± 0.52 A	11.43 ± 0.263 B	12.46 ± 0.82 C	5	<0.01
	(4)	IA (°)	31.2 ± 1.93 A	29.5 ± 1.83 B	27.2 ± 1.59 C	5	<0.01
	(5)	Moment (kNm)[a]	9.33 ± 0.48 A	10.61 ± 0.86 B	11.87 ± 0.62 C	5	<0.01
T3	(6)	MOM (kNm)	8.36 ± 0.42 A	7.83 ± 0.57 B	7.42 ± 0.68 C	5	<0.01
	(7)	IA (°)	26.7 ± 1.73 A	24.5 ± 1.61 B	22.2 ± 1.49 C	5	<0.01

Data are means ± 1 S.E.

[a] Moment indicates the moment in T2 caused by the inclination angle was similar to T1. Within rows, values with different letters are significantly different at 0.01 levels.

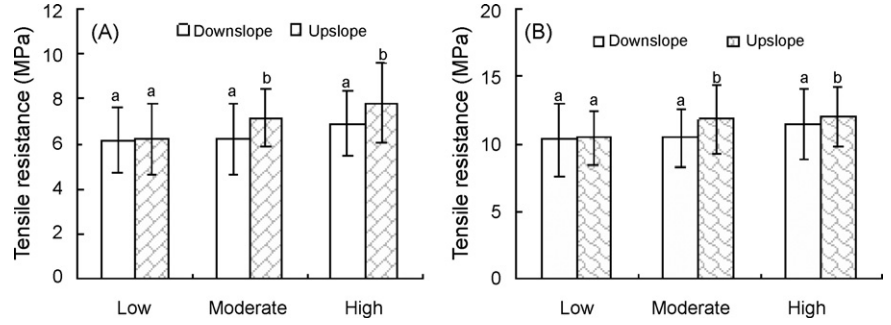


Fig. 5 – Comparison of the tensile resistance between the upslope and downslope. (A) and (B) The tensile resistance of the first- and second-order lateral roots, respectively. Data are means ± 1 S.E., $n = 10$ plants per slope. For each parameter, means with the same letter are not significantly ($P > 0.05$) difference.

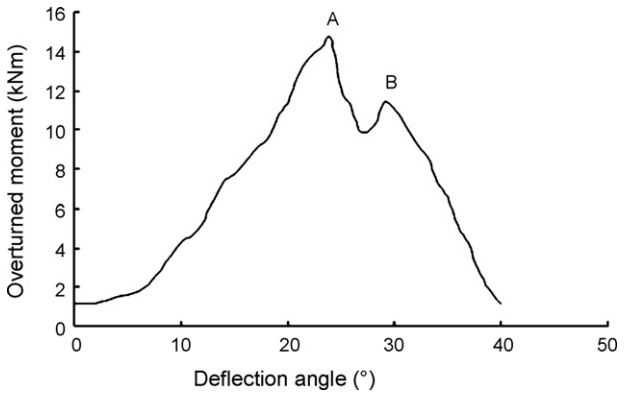


Fig. 6 – Relationship between overturning moment and inclination angle of trunks in Test 1. The line represents one of winched trees on the medium slope (with plant height of 11.23 m and BD of 10.11 cm). Two letters of A and B indicated the maximal and the second peak value.

remarked higher than ones of the other two types of slopes ($P < 0.01$), while the inclination angles of the winched plants for the high slope was significant smaller than the moderate and low slopes ($P < 0.01$).

The inclination angles in both the T1 and T3, showed no difference between and among all treatments (all $P \geq 0.64$), but those inclination angles were smaller than ones of all treatments in T2 ($P < 0.01$).

All first-order lateral roots on downslope side in T1 and T3 were broken, and their breakage points were about 20 cm from the trunk. In T1 and T2, the root systems, including the first- and second-order lateral roots, failed as winched plants were pulled down, but the roots that hold in soil in upslope were only the finest tips. All taproots were pulled out when the inclination angle of the trunks ranged 40–50°, while the roots in soil were only fiber roots.

In T1, the overturning moment initially increased gradually with displacement to a peak value, then dropped dramatically, and subsequently increased to the second peak (lower than the former), before lastly declining gradually (Fig. 6). The patterns that change in overturning moments with displacement among different slopes were the same nearly.

4. Discussion

4.1. *Distribution of root system*

Environmental factors, such as soil nutrient and water condition, may lead to the varied patterns of root distribution (Coutts, 1987; Canadell et al., 1996). In our present study, difference in nutrient and water conditions among three slopes may be ignored (all $P > 0.12$) (Table 3) due to selected plants under artificial management. Therefore, difference of root systems distribution should be attributed to mechanical simulation such as wind and tree self-loading (Jaffe, 1973; Gartner, 1994).

The plants growing on the high slopes often develop a larger first-order lateral root but with fewer number, while those on the low slope develop a larger number of first-order lateral roots. Di Iorio et al. (2005) suggested that the pattern of roots distribution on flat land may be attribute to weaker turning moment induced by the trunk. Moreover, many previous studies demonstrated that the roots of mechanical stressed plants were thicker and with more number (Goodman and Ennos, 1996, 1997; Mickovski and Ennos, 2003). The roots growing in the upslope, both the first- and second-order lateral develop better than ones in downslope, suggesting that the upslope-grown roots may be affected by mechanical stress, and this is consistent with previous report that plants' tigmomorphgenesis could serve to improve its anchorage when mechanical stress affects plant stability (Jaffe and Biro, 1979).

The many previous studies demonstrated that the degree of deformation of roots growing on steep slope were greater than that of flat-grown, and upslope-grown roots were stronger than downslope-grown in several plant species (Chiatante et al., 2003; Di Iorio et al., 2005) owing to asymmetry distribution of root system induced by wind loading, self-loading and soil self-loading. Furthermore, the number of the roots growing in downslope side was much less than that in upslope (Di Iorio et al., 2005). In this study, for the moderate and high slopes, differences in both amount and diameter between the upslope- and downslope-grown roots were found, which are similar to previous studies that the numbers of the first- and second-order lateral roots growing on upslope side were larger (Di Iorio et al., 2005). Wind loading could be excluded under present study condition, and the distribution and architecture of root systems were mainly the result from plant self-loading and soil self-loading induced by slope gradient. Moreover, the roots of low slope-grown have no difference between the upslope and downslope and this might be due to a smaller mechanical stimulation under low slope.

4.2. *Tensile strength of root system*

Apart from roots' morphological features, the manner/strategy that root systems improvement anchorage is to reinforce its properties such as strength, stiffness when mechanical stimulation (Stokes and Guitard, 1997; Nicoll, 2000; Niklas et al., 2000). The upslope-grown roots both in the moderate and high slope were subjected to mechanical stress (such as soil weight, plant own weight, etc.), this stress in turn trends to stimulate improvement tensile strength of the roots. Therefore, the tensile strength of the upslope-grown roots in those two slopes was higher than that of low slope-grown. Similarly, the tensile strength of upslope-grown roots within the same slope was also higher than that of downslope-grown. A result agrees with previous report by Chiatante et al. (2003).

It has been suggested that fluctuations in tensile strength (Hathaway and Penny, 1975) might be related to variation in ratio of the lignin/cellulose, which is under seasonal and/or abiotic factors such as mechanical stress (Plomìon et al., 2001). For the moderate and high slope, the differences in tensile strength between upslope- and downslope-grown roots might also be related to different ratio of the lignin to cellulose. To demonstrate this, further study may be needed to understand the relationship between tensile strength and lignin content of root.

4.3. *Anchorage and breakage of root system*

During the process of winched plants turning over toward downslope, the roles of roots growing on the upslope and downslope were different completely. The upslope-grown roots can service to resist uprooting (or breakage), while the downslope roots trend to resist bending (Crook and Ennos, 1996). Contributions of roots growing on different slopes and directions to plant anchorage show a considerable difference. The contributions of the upslope-grown roots on the low, moderate and high slopes for the anchorage reached 37%, 44% and 50%, respectively, while the contributions of the downslope-grown on those three slopes were 30%, 25% and 21%, respectively. These results indicated that the contributions of different directional roots to plant anchorage were closely related to slope gradient.

The downslope lateral roots were nearly broken due to bending resistance during the winched trees overturning, similar phenomenon was also observed on larch anchorage by Crook and Ennos (1996). In general, it was assumed that roots crossing a shear zone generate tensile strength, are elongate in tension, and break at the tips, not in the shear zone (Di Iorio et al., 2005). Moreover, most of the upslope-grown roots were pulled out during a pull-out test leaving the finest distal portions still embedded in soil. The mode of root failure is similar to that occurring during a pull-out test (O'Loughlin and Watson, 1979; Abe and Ziemer, 1991a, b). Whole taproots were pulled out completely, which may be attributed to higher soil moisture (Goodman and Ennos, 1999).

Change in the overturning moments with displacement might be related to the breakage of different directional root systems. The inclination angles in both T 1 and T3 were nearly the same as overturning moment reached a maximum value, probably indicating that the overturning moment declining firstly was contributed by the failure of downslope-grown roots (point A in Fig. 6). Moreover, the inclination angles in T1 created by overturning moment that reached the second peak were similar to one in T2 created by the maximum overturning moment, may suggest that the second decline of the overturning moment (point B in Fig. 6) was linked to the roots failure on upslope side.

5. Conclusion

Slope gradient had an important impact on distribution and anchorage resistance of root systems of *P. yunnanensis*. Anchorage resistance on the high slope was markedly higher than that on moderate and low slopes. The roots growing on upslope side were mainly contributors for plant anchorage, and the share they contribute increases with slops gradient. Plant roots can respond to mechanical stimulation, and the strength, length and amount for upslope-grown roots, including the first and second lateral, could increase in order to offset plant self-loading as slope increases. So that both plant and slope stability could better improve. These findings suggest that the architecture and distribution of root systems can impact on the plant anchorage. It is also useful to guide how to select species for vegetation restoration in ecological engineering. A lateral root well-developed, dwarfish woody species could be in favor of slope stability.

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